

Least Squares Regression

Example 04-05: Find least squares regression using polynomials of order 1 for the following data.

i	x_i	y_i
0	1	0.5
1	2	2.5
2	3	2.0
3	4	4.0
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6	7	5.5

$$y(x) = P_1(x, \mathbf{a}) = a_0 + a_1x$$

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$n = 6$

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$$a_0 \cdot \sum_{i=0}^6 1 + a_1 \cdot \sum_{i=0}^6 x_i = \sum_{i=0}^6 y_i$$

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$$7a_0 + 28a_1 = 24$$

$$\frac{\partial S_e}{\partial a_1} = \sum_{i=0}^6 2 \cdot (y_i - a_0 - a_1x_i) \cdot (-x_i) = 0$$

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$$a_0 = 0.0714286, a_1 = 0.8392857$$

$$P_1(x) = 0.0714286 + 0.8392857x$$

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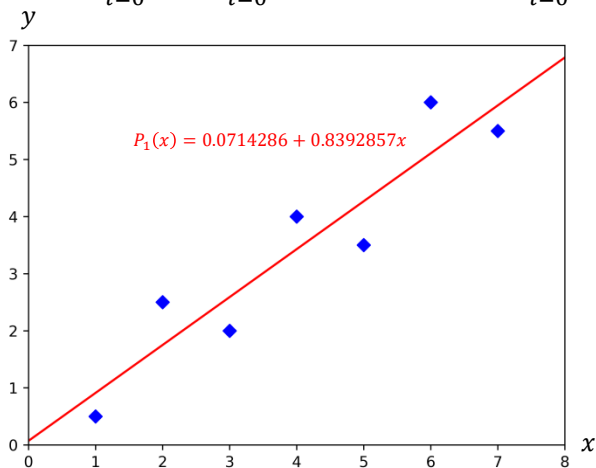
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Example 04-06: Find least squares regression using polynomials of order 2 for the following data.

i	x_i	y_i
0	0.00	1.0000
1	0.25	1.2840
2	0.50	1.6487
3	0.75	2.1170
4	1.00	2.7183

$$y(x) = P_2(x, \mathbf{a}) = a_0 + a_1x + a_2x^2$$

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$$\frac{\partial S_e}{\partial a_0} = \sum_{i=0}^4 2 \cdot (y_i - a_0 - a_1x_i - a_2x_i^2) \cdot (-1) = 0 \qquad 5a_0 + 2.5a_1 + 1.875a_2 = 8.768$$

$$\frac{\partial S_e}{\partial a_1} = \sum_{i=0}^4 2 \cdot (y_i - a_0 - a_1x_i - a_2x_i^2) \cdot (-x_i) = 0 \qquad 2.5a_0 + 1.875a_1 + 1.5625a_2 = 5.4514$$

$$\frac{\partial S_e}{\partial a_2} = \sum_{i=0}^4 2 \cdot (y_i - a_0 - a_1x_i - a_2x_i^2) \cdot (-x_i^2) = 0 \qquad 1.875a_0 + 1.5625a_1 + 1.3828125a_2 = 4.4015375$$

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$$1.875a_0 + 1.5625a_1 + 1.3828125a_2 = 4.4015375$$

$$a_0 = 1.0051371, a_1 = 0.8641827, a_2 = 0.8436571$$

$$P_2(x) = 1.0051371 + 0.8641827x + 0.8436571x^2$$

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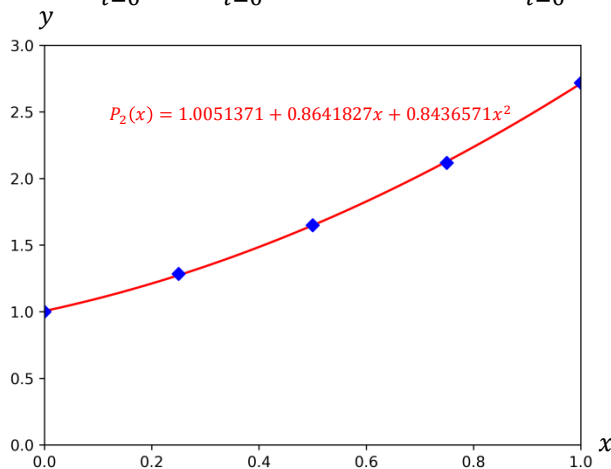
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$$5a_0 + 2.5a_1 + 1.875a_2 = 8.768$$

$$2.5a_0 + 1.875a_1 + 1.5625a_2 = 5.4514$$

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Example 04-07: Find least squares regression using polynomials of order 1 passing origin for the following data.

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$$e_i = y_i - P_1(x_i, \mathbf{a}) = y_i - a_1x_i$$

$$S_e = \sum_{i=0}^6 e_i^2 = \sum_{i=0}^6 [y_i - P_1(x_i, \mathbf{a})]^2 = \sum_{i=0}^6 (y_i - a_1x_i)^2$$

$$\frac{\partial S_e}{\partial a_1} = \sum_{i=0}^6 2 \cdot (y_i - a_1x_i) \cdot (-x_i) = 0$$

$$a_1 \cdot \sum_{i=0}^6 x_i^2 = \sum_{i=0}^6 x_i y_i$$

$$a_1 = \frac{119.5}{140} = 0.8535714$$

$$P_1(x) = 0.8535714x$$

Least Squares Regression

Example 04-07: Find least squares regression using polynomials of order 1 passing origin for the following data.

$n = 6$

i	x_i	y_i
0	1	0.5
1	2	2.5
2	3	2.0
3	4	4.0
4	5	3.5
5	6	6.0
6	7	5.5

$$y(x) = P_1(x, \mathbf{a}) = a_1x \quad (a_0 = 0, m = 1)$$

$$e_i = y_i - P_1(x_i, \mathbf{a}) = y_i - a_1x_i$$

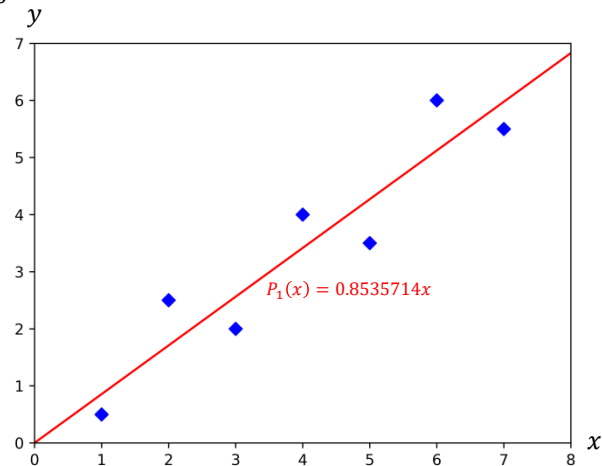
$$S_e = \sum_{i=0}^6 e_i^2 = \sum_{i=0}^6 [y_i - P_1(x_i, \mathbf{a})]^2 = \sum_{i=0}^6 (y_i - a_1x_i)^2$$

$$\frac{\partial S_e}{\partial a_1} = \sum_{i=0}^6 2 \cdot (y_i - a_1x_i) \cdot (-x_i) = 0$$

$$a_1 \cdot \sum_{i=0}^6 x_i^2 = \sum_{i=0}^6 x_i y_i$$

$$a_1 = \frac{119.5}{140} = 0.8535714$$

$$P_1(x) = 0.8535714x$$



Assignment 3

Consider the following data sets.

Data Set 1		Data Set 2	
x_i	$f(x_i)$ or y_i	x_i	$f(x_i)$ or y_i
1.0	1.84	0.00	1.000
1.1	1.96	0.15	1.004
1.3	2.21	0.31	1.031
1.5	2.45	0.50	1.117

1. Find Lagrange interpolating polynomials of order 3 for each data set.
2. Find least squares regression for Data Set 1 using the following equations
 - 2.1 $P_1 = a_0 + a_1x$
 - 2.2 $P_2 = a_0 + a_1x + a_2x^2$
3. Find least squares regression for Data Set 2 using the following equations
 - 3.1 $P_1 = 1 + a_1x$
 - 3.2 $P_2 = 1 + a_1x + a_2x^2$